IEOR E4004 — Optimization Models and Methods

Project I — Childcare Desert Reduction in New York City

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This project studies how New York City can reduce childcare deserts at minimum cost by combining expansions of existing childcare facilities with the opening of new facilities. New York City classifies a zip code as a childcare desert when licensed capacity is too low relative to local child population.

In high-demand areas, defined as zip codes with employment rate at least 60% or average income at most $60,000, the relevant threshold is one half of the population aged 0 to 12; otherwise it is one third. NYC also imposes a separate requirement for children aged 0 to 5: available under-5 slots should be at least two thirds of the local 0 to 5 population [1].

This project studies the minimum public funding needed to eliminate coded childcare deserts across NYC. We solve the problem in two stages: Part 1 builds an implemented benchmark model that allows free siting of new facilities and more flexible expansion of existing ones. Part 2 keeps the same demand logic but imposes tighter expansion caps, candidate-site restrictions, and spacing rules, producing a more operational planning model.

Code availability. The working code is organized as a seven-stage notebook pipeline in the submitted repository: shared data cleaning, benchmark parameter estimation, benchmark optimization, benchmark visualization, realistic parameter estimation, realistic optimization and validation, and final comparison visualization. The workflow first constructs shared inputs, then solves Part 1, then solves Part 2, and finally exports the tables and figures used in this report. The submitted repository contains the notebook pipeline under `notebooks/` and the report files under `report/`. Figure 1 gives a compact schema of that pipeline.

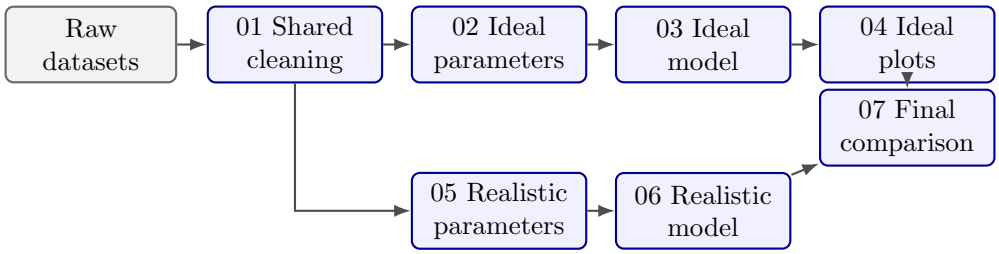

Figure 1: Notebook pipeline used to construct shared inputs, solve both models, and export report figures.

1 Background and Problem Framing

Childcare deserts matter because lack of access affects both labor-force participation and early-childhood care, a point emphasized in the broader childcare-desert literature summarized by the Center for American Progress [3]. The project framing here is operational rather than behavioral: demand is enforced at the zip-code level, and the government decides how much to spend on two levers, expanding existing facilities and opening new ones. That makes the problem well suited to mixed-integer optimization, but it also means the results should be interpreted as planning solutions rather than household travel models.

The two-part structure is useful because the questions are different. Part 1 asks what the cost would be under optimistic implementation conditions. Part 2 asks how that answer changes when the city loses the ability to build anywhere, must respect site spacing, and can only expand incumbent facilities modestly.

2 Parameter Estimation

2.1 Shared data cleaning

The notebook pipeline uses five raw datasets supplied with the assignment: regulated childcare facilities, potential locations, employment rates, average individual income, and child population by zip code. The shared-cleaning stage standardizes zip codes to five-character strings, harmonizes column names, converts numeric fields, and preserves facilities even when geographic coordinates are missing. The cleaned shared outputs are then reused by both models.

Table 1 shows the cleaned shared tables passed from Notebook 01 to the later parameter-estimation and optimization stages.

Shared output	**Level**	**Shape**	**Used for**
Clean facility table	facility	$7{,}740 \times 20$	Standardized facility master table with active flag, geo flag, under-5 proxy, and quality flags
Facility geo-ready table	facility	$7{,}740 \times 16$	Compact facility subset used for realistic distance screening
Clean population table	zipcode	211×22	Harmonized age bands plus derived child-population fields
Clean employment table	zipcode	179×2	Zipcode employment covariate before imputation
Clean income table	zipcode	179×2	Zipcode income covariate before imputation
Candidate geo-ready table	candidate	$31{,}100 \times 5$	Standardized candidate ID, zip code, coordinates, and geo flag
Master zipcode universe	zipcode	311×12	Merge backbone with source flags, facility counts, baseline capacity, and candidate counts

Table 1: Shared cleaned outputs generated by Notebook 01

Three modeling choices in the shared pipeline materially affect later steps. First, active facilities with positive licensed capacity define the incumbent supply universe. Second, `children_capacity` is used as the under-5 capacity proxy for existing facilities. Third, missing employment and income values are imputed with citywide medians, while missing child-population entries are carried as zero after the merged zip-code table is formed. The processed ideal inputs cover 311 zip codes and 7,712 active facilities.

2.2 Demand- and supply-side parameters

For each zip code z, the benchmark stage defines the high-demand indicator as

$$\text{high_demand}_z = \mathbf{1}\{\text{employment}_z \geq 0.60 \text{ or income}_z \leq 60{,}000\}.$$

The corresponding total-capacity requirement is

$$\text{req_total}_z = \begin{cases} \lfloor 0.5 \cdot \text{pop}_{0\text{–}12,z} \rfloor + 1, & \text{high demand}, \\ \lfloor \frac{1}{3} \cdot \text{pop}_{0\text{–}12,z} \rfloor + 1, & \text{otherwise}, \end{cases}$$

and the under-5 requirement is

$$\text{req_under5}_z = \left\lceil \frac{2}{3} \text{pop}_{0\text{–}5,z} \right\rceil.$$

When observed child population is zero, the implemented pipeline sets $\text{req_total}_z = 0$ and does not classify the zip code as a baseline total-capacity desert. Existing supply is aggregated from facility-level licensed capacity and under-5 proxy capacity within each zip code.

2.3 Intervention parameters

New-facility options are unchanged across both parts and come directly from the assignment.

Facility type	Total slots	Under-5 max	Fixed cost ($)
Small	100	50	65,000
Medium	200	100	95,000
Large	400	200	115,000

Table 2: New-facility build menu used in both models

Part 1 benchmark parameters. The instructor-approved expansion-cap interpretation used in the notebook is

$$x_f \le \min(1.2n_f,\ 500).$$

Because the assignment does not specify the first-block cost function $c_{\text{slots}}(n_f)$, the benchmark parameter-estimation stage uses

$$c_f^{(1)} = \begin{cases} 350, & n_f < 25, \\ 300, & 25 \le n_f < 75, \\ 250, & n_f \ge 75, \end{cases} \qquad \text{dollars per added slot,}$$

and, for expansions at or above current capacity,

$$c_f^{(2)} = \frac{20000 + 200\,n_f}{n_f}.$$

The implemented Part 1 model then prices each facility's expansion at $c_f^{(1)}$ if $x_f < n_f$ and at $c_f^{(2)}$ if $x_f \ge n_f$.

Part 2 realistic parameters. The realistic parameter-estimation stage replaces the single benchmark cap with three cumulative-floor expansion blocks:

$$\text{cap1}_f = \lfloor 0.10\,n_f \rfloor, \qquad \text{cap2}_f = \lfloor 0.15\,n_f \rfloor - \lfloor 0.10\,n_f \rfloor, \qquad \text{cap3}_f = \lfloor 0.20\,n_f \rfloor - \lfloor 0.15\,n_f \rfloor.$$

These correspond to the 0% to 10%, 10% to 15%, and 15% to 20% expansion bands, so total realistic expansion room at facility f is $\lfloor 0.20\,n_f \rfloor$. The associated block-cost coefficients are

$$\text{coef1}_f = \frac{20000 + 200\,n_f}{n_f}, \qquad \text{coef2}_f = \frac{20000 + 400\,n_f}{n_f}, \qquad \text{coef3}_f = \frac{20000 + 1000\,n_f}{n_f}.$$

The processed inputs imply a citywide realistic expansion room of 61,197 slots. The shared benchmark demand table also carries the zero-child rule into both models, so zip codes with zero observed child population have zero total-slot requirement.

Screening metric	**Value**
Candidate sites examined	31,100
Blocked candidate sites	2,578
Feasible candidate sites	28,522
Candidate-candidate conflict pairs	11,455
Candidate-existing conflict pairs	4,730
Existing-existing close pairs (diagnostic only)	7,644
Facilities missing coordinates	169

Table 3: Part 2 site-screening outputs from the realistic parameter-estimation stage

3 Methods

3.1 Part 1 benchmark model

The benchmark model is defined over zip codes $z \in Z$, active existing facilities $f \in F$, and build sizes $s \in S = \{\text{small}, \text{medium}, \text{large}\}$. The key decision variables are

$$\begin{aligned}
x_f &= \text{total expansion slots at facility } f, \\
u_f &= \text{under-5 slots assigned within expansion}, \\
\delta_f &= \text{binary indicator for the high-cost expansion regime}, \\
w_f &= \text{auxiliary facility expansion-cost variable}, \\
y_{zs} &= \text{number of new facilities of size } s \text{ in zip } z, \\
g_{zs} &= \text{under-5 slots assigned within those new builds.}
\end{aligned}$$

The objective is to minimize modeled expansion cost, fixed build cost, and \$100 for each newly created under-5 slot:

$$\min \sum_{f \in F} w_f + \sum_{z \in Z} \sum_{s \in S} K_s y_{zs} + 100 \left(\sum_{f \in F} u_f + \sum_{z \in Z} \sum_{s \in S} g_{zs} \right).$$

The core constraints enforce facility-level expansion caps, under-5 assignment limits, threshold logic for the two cost regimes, and zip-code service requirements:

$$x_f \le U_f, \qquad u_f \le x_f, \qquad x_f \le (n_f - 1) + U_f \delta_f, \qquad x_f \ge n_f \delta_f,$$

$$\text{existing_total}_z + \sum_{f \in F(z)} x_f + \sum_{s \in S} B_s y_{zs} \ge \text{req_total}_z,$$

$$\text{existing_under5}_z + \sum_{f \in F(z)} u_f + \sum_{s \in S} g_{zs} \ge \text{req_under5}_z, \qquad g_{zs} \le U_s y_{zs}.$$

Big-M linking constraints then force $w_f = c_f^{(1)} x_f$ when $\delta_f = 0$ and $w_f = c_f^{(2)} x_f$ when $\delta_f = 1$, so the whole expansion is priced under one regime.

This model is intentionally optimistic. It does not represent siting feasibility, minimum spacing, or cross-zip travel. It is best interpreted as the implemented benchmark model rather than a deployable plan.

3.2 Part 2 realistic model

Part 2 keeps the same demand targets but changes the intervention structure. Existing facilities may expand only within three capped bands, and new facilities can only be opened on observed candidate sites $l \in L$. The key parameters are the three incumbent expansion caps $\text{cap1}_f, \text{cap2}_f, \text{cap3}_f$, the corresponding coefficients $\text{coef1}_f, \text{coef2}_f, \text{coef3}_f$, the blocked-site set L_{blocked}, the within-zipcode conflict set $\mathcal{C}$, and the build-menu parameters K_s, B_s, U_s. The key decision variables are

$$\begin{aligned} x_{1f}, x_{2f}, x_{3f} &= \text{added slots in the 0\%–10\%, 10\%–15\%, 15\%–20\% bands,} \\ u_f &= \text{under-5 slots assigned within incumbent expansion,} \\ y_{ls} &= \text{binary indicator for opening size } s \text{ at candidate site } l, \\ g_{ls} &= \text{under-5 slots assigned within that selected facility.} \end{aligned}$$

The implemented objective is

$$\min \sum_{f \in F} (\text{coef1}_f x_{1f} + \text{coef2}_f x_{2f} + \text{coef3}_f x_{3f}) + \sum_{l \in L} \sum_{s \in S} K_s y_{ls} + 100 \left(\sum_{f \in F} u_f + \sum_{l \in L} \sum_{s \in S} g_{ls} \right).$$

The main feasibility constraints are

$$x_{1f} \le \text{cap1}_f, \qquad x_{2f} \le \text{cap2}_f, \qquad x_{3f} \le \text{cap3}_f, \qquad u_f \le x_{1f} + x_{2f} + x_{3f},$$

$$\sum_{s \in S} y_{ls} \le 1, \qquad g_{ls} \le U_s y_{ls},$$

$$\sum_{s \in S} y_{ls} = 0 \quad \forall l \in L_{\text{blocked}}, \qquad \sum_{s \in S} y_{l_1 s} + \sum_{s \in S} y_{l_2 s} \le 1 \quad \forall (l_1, l_2) \in \mathcal{C},$$

$$\text{existing_total}_z + \sum_{f \in F(z)} (x_{1f} + x_{2f} + x_{3f}) + \sum_{(l,s) \in L(z)} B_s y_{ls} \ge \text{req_total}_z,$$

$$\text{existing_under5}_z + \sum_{f \in F(z)} u_f + \sum_{(l,s) \in L(z)} g_{ls} \ge \text{req_under5}_z.$$

These choices match the NYC planning context described in the assignment. Banded incumbent variables make the tighter 20% cap and increasing marginal expansion difficulty transparent, while candidate-site binary variables make the 0.06-mile screening rule enforceable. Because existing facilities are fixed assets in the current planning horizon, existing-existing proximity issues are tracked as diagnostics rather than modeled as closures or relocations.

Implemented Part 2 model. The realistic optimization stage solves a citywide MILP with 201,980 variables and 155,819 constraints. The model enforces the coded site-screening and service constraints, but the expansion-cost objective is an additive increasing-cost approximation rather than the assignment's literal piecewise-total formula. Existing-existing distance conflicts are recorded diagnostically, not enforced.

4 Results

4.1 Part 1 benchmark results

The benchmark optimization stage solves the Part 1 model to proven optimality in 3.73 seconds. The implemented benchmark objective is $126.67 million, split across expansion, new construction, and under-5 equipment.

Benchmark takeaway. The implemented benchmark requires $126.67 million, including 206,364 expansion slots, 386 new facilities, and 135 facilities that cross into the higher-cost expansion regime. Even under optimistic siting assumptions, new construction is still necessary.

Part 1 metric	**Value**
Total objective ($)	126,668,941
Expansion cost ($)	54,716,841
New-build cost ($)	44,180,000
Under-5 equipment cost ($)	27,772,100
Added expansion slots	206,364
Added under-5 expansion slots	202,938
New facilities built	386
Added new-build slots	152,900
Added under-5 new-build slots	74,783
Zip codes meeting all coded requirements after solve	311

Table 4: Implemented Part 1 benchmark results

4.2 Part 2 realistic results

The realistic optimization stage solves the Part 2 model in 18.21 seconds with a final MIP gap below 0.1%. The objective rises to $183.96 million, and the intervention mix shifts sharply toward new construction.

Realistic-model takeaway. The implemented realistic model costs $183.96 million. Expansion falls to 25,108 slots, while new construction rises to 1,284 facilities and 507,700 new slots.

Part 2 metric	**Value**
Total objective ($)	183,963,207
Expansion cost ($)	9,381,107
New-build cost ($)	146,810,000
Under-5 equipment cost ($)	27,772,100
Expanded facilities	1,126
Added expansion slots	25,108
Added under-5 expansion slots	25,067
New facilities built	1,284
Added new-build slots	507,700
Added under-5 new-build slots	252,654
Zip codes with zero under-5 slack	311
Zip codes meeting all coded requirements after solve	311

Table 5: Implemented Part 2 realistic results

Validation checks. The exported validation table from the realistic optimization stage reports 311/311 zip codes meeting both service requirements, 0 blocked candidate sites selected, 0 candidate-conflict violations, 0 facility cap violations, and 0 new builds in zero-child zip codes.

4.3 Comparison and interpretation

Metric	**Part 1**	**Part 2**	**Change**
Total cost ($)	126,668,941	183,963,207	+57,294,267
Expansion slots	206,364	25,108	-181,256
New-build slots	152,900	507,700	+354,800
New facilities	386	1,284	+898
Under-5 equipment cost ($)	27,772,100	27,772,100	0

Table 6: Benchmark versus realistic intervention mix

Main shift from Part 1 to Part 2. Realistic restrictions add $57.29 million to the objective and move the solution away from incumbent expansion toward many more large new facilities. The under-5 requirement remains the tightest system-wide driver in both models.

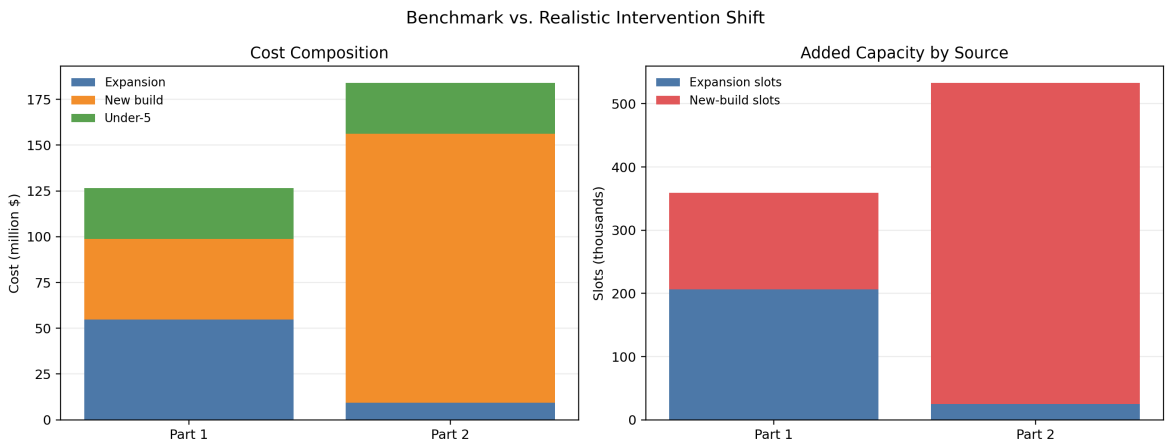

Figure 2: Cost and capacity composition shift from Part 1 to Part 2

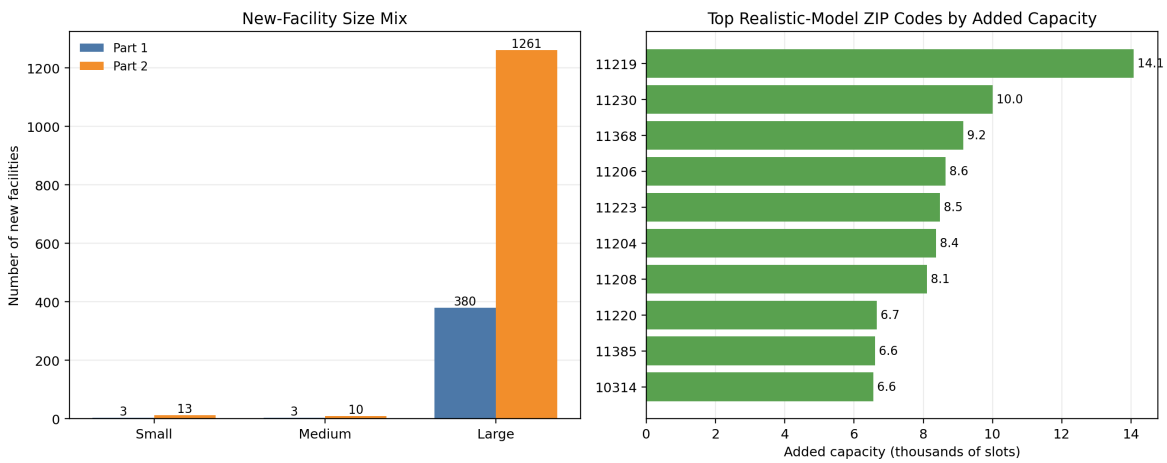

Figure 3: Part 2 size mix and top zip codes by added capacity

Two patterns stand out. First, the realistic solution is dominated by large new facilities: 1,261 of the 1,284 selected new builds are large. Second, the under-5 service requirement binds everywhere in the final realistic solution: all 311 zip codes have zero under-5 slack, while only 130 have zero total-capacity slack. The model therefore creates visible total-capacity oversupply in order to meet the under-5 requirement with the available build menu.

The highest-intervention zip codes remain concentrated in a relatively small set of areas, including 11219, 11230, 11368, 11206, 11223, and 11204. In those locations, limited incumbent expansion room and strong under-5 demand force new construction rather than marginal adjustments to existing facilities.

5 Conclusions

The seven-stage notebook pipeline supports a clear benchmark-to-realistic comparison. Under the implemented benchmark model, NYC can eliminate coded childcare deserts for $126.67 million. Under the implemented realistic model, the cost rises to $183.96 million because the city loses flexible siting and can expand incumbent facilities only modestly.

The key policy insight is not just that the realistic model is more expensive. It is that the intervention mechanism changes. Once expansion becomes tightly capped and candidate-site spacing is enforced, the city must fund many more large new facilities. Across both parts, the under-5 requirement is the dominant planning driver.

Bibliography

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[3] Center for American Progress. *Child Care Deserts in the United States*. https://www.americanprogress.org/series/child-care-deserts/.